Nour Tamer Alhalwagy

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Education

Faculty of Computer Science and Artificial Intelligence. Major: Information Security and Digital Forensics, 2023 – 2027
Benha University Egypt

Experience

DEPI - Digital Egypt Pioneers Initiative, *AI & Data Science - Microsoft Machine Learning Engineer Trainee* 11/2025 – 07/2026
Cairo-Egypt

- Built solid foundation in Data Science and AI development, applying Machine Learning, Deep Learning, NLP, and Computer Vision techniques.
- Worked with Azure AI services and MLOps pipelines to develop, manage, and deploy machine learning solutions.
- Collaborated in agile environments and practiced clean code principles using python on diverse structured datasets, using algorithm selection, hyperparameter tuning, and model evaluation techniques to deliver robust outcomes.
- Implemented and optimized machine learning models for classification and forecasting tasks across multiple use cases, improving predictive reliability through rigorous validation and performance analysis.
- Delivered multiple end-to-end machine learning solutions across diverse business domains, applying appropriate algorithms to achieve target performance thresholds.

GUPCO - Gulf of Suez Petroleum, *IT Intern* 08/2025 – 08/2025
Maadi - Egypt

- Gained knowledge of Agile workflows, including planning, iterations, and continuous improvement.
- Developed a strong foundation in AI, Machine Learning, and Deep Learning concepts.
- Expanded knowledge of modern frameworks, software development, and applications.
- Delivered a technical presentation to IT managers on Artificial Intelligence and Machine Learning, explaining key concepts, use cases, and business impact.

VOIS - Vodafone Intelligent Solutions, *Technical Support Specialist* (*Full-Time*) 05/2024 – 11/2024
Smart Village - Egypt

- Monitored and assessed network coverage and performance across different areas.
- Performed troubleshooting for data roaming and 5G data issues for UK customers.
- Diagnosed connectivity problems and provided first-level technical support.
- Escalated complex or large-scale issues to higher technical teams for resolution.

Volunteering and extracurricular

Google Developer Groups (On Campus BU), *Vice-Head of AI & Data Science* 05/2026 – Present

- Led training programs covering Python, NLP (BERT, Transformers), ML/DL (PyTorch, TensorFlow), and deployment (FastAPI, Flask) for 300+ members
- Managed team of 40+ data scientists, overseeing 25+ projects from EDA to production deployment
- Designed curriculum and hosted 95+ technical sessions (workshops/bootcamps) with 90%+ satisfaction rate
- Implemented assessment system to track member progress through hands-on projects and code reviews

IMF-International Monetary Fund, *Delegate* Washington - US

- Attended sessions on fiscal policy for inclusive GenAI growth and the future of cross-border payments, focusing on digital tokens and programmable platforms.
- learnt more about real crisis solved by AI and its real-world impact on economies, businesses and global systems.

Skills

Programming languages: — Python, SQL, C++
Software Engineering — OOP, Data Structure and Algorithms
ML/DL Frameworks: — JAX, PyTorch, TensorFlow, Keras; Supervised/Unsupervised Learning;
Deployment & MLOps: — FastAPI, Flask, Streamlit, Docker, Git, CI/CD, Microsoft Azure

Projects

Smart AI Driving Alert Present

- Developed a demo AI-based driving alert system for traffic penalty avoidance.
- Technologies: Python, Overpass API, HTML, JSON, NumPy, Pandas, Jupyter Notebook.

Autonomous Drone 06/2026

- Trained an autonomous drone using Deep Reinforcement Learning (DQN/PPO) in the AirSim simulator.
- Integrated YOLOv8-based computer vision for real-time object detection and target localization.
- Implemented state representation and action selection for autonomous navigation.
- Designed and optimized the reward function to enable autonomous tracking and navigation toward a designated target vehicle.

Checkers (Human Vs AI) 04/2026

- Fully functional Checkers game in Python using Pygame.
- Implemented Minimax algorithm with Alpha-Beta pruning (depth 3). Technologies: Python, Pygame, Minimax, Alpha-Beta Pruning

Certificates

- Microsoft Certified ML engineer
- Mathematics for Machine Learning from DeepLearning.AI
- AWS AI Practitioner from Udacity
- Data analysis from Udacity X FWD